

$$f(x) = \begin{cases} \frac{1}{36}(9 - x^2) & \text{for } -3 \leq x \leq 3, \\ 0 & \text{otherwise.} \end{cases}$$

Sketch this p.d.f. and determine the values of the following probabilities: **a.** $\Pr(X < 0)$ **b.** $\Pr(-1 \leq X \leq 1)$ **c.** $\Pr(X > 2)$.

4. Suppose that the p.d.f. of a random variable X is as follows:

$$f(x) = \begin{cases} cx^2 & \text{for } 1 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

- a.** Find the value of the constant c and sketch the p.d.f.
- b.** Find the value of $\Pr(X > 3/2)$.

5. Suppose that the p.d.f. of a random variable X is as follows:

$$f(x) = \begin{cases} \frac{1}{8}x & \text{for } 0 \leq x \leq 4, \\ 0 & \text{otherwise.} \end{cases}$$

- a.** Find the value of t such that $\Pr(X \leq t) = 1/4$.
- b.** Find the value of t such that $\Pr(X \geq t) = 1/2$.

6. Let X be a random variable for which the p.d.f. is as given in Exercise 5. After the value of X has been observed, let Y be the integer closest to X . Find the p.f. of the random variable Y .

7. Suppose that a random variable X has the uniform distribution on the interval $[-2, 8]$. Find the p.d.f. of X and the value of $\Pr(0 < X < 7)$.

8. Suppose that the p.d.f. of a random variable X is as follows:

$$f(x) = \begin{cases} ce^{-2x} & \text{for } x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

- a.** Find the value of the constant c and sketch the p.d.f.
- b.** Find the value of $\Pr(1 < X < 2)$.

9. Show that there does not exist any number c such that the following function $f(x)$ would be a p.d.f.:

$$f(x) = \begin{cases} \frac{c}{1+x} & \text{for } x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

10. Suppose that the p.d.f. of a random variable X is as follows:

$$f(x) = \begin{cases} \frac{c}{(1-x)^{1/2}} & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

- a.** Find the value of the constant c and sketch the p.d.f.
- b.** Find the value of $\Pr(X \leq 1/2)$.

11. Show that there does not exist any number c such that the following function $f(x)$ would be a p.d.f.:

$$f(x) = \begin{cases} \frac{c}{x} & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

12. In Example 3.1.3 on page 94, determine the distribution of the random variable Y , the electricity demand. Also, find $\Pr(Y < 50)$.

13. An ice cream seller takes 20 gallons of ice cream in her truck each day. Let X stand for the number of gallons that she sells. The probability is 0.1 that $X = 20$. If she doesn't sell all 20 gallons, the distribution of X follows a continuous distribution with a p.d.f. of the form

$$f(x) = \begin{cases} cx & \text{for } 0 < x < 20, \\ 0 & \text{otherwise,} \end{cases}$$

where c is a constant that makes $\Pr(X < 20) = 0.9$. Find the constant c so that $\Pr(X < 20) = 0.9$ as described above.

3.3 The Cumulative Distribution Function

Although a discrete distribution is characterized by its p.f. and a continuous distribution is characterized by its p.d.f., every distribution has a common characterization through its (cumulative) distribution function (c.d.f.). The inverse of the c.d.f. is called the quantile function, and it is useful for indicating where the probability is located in a distribution.

Example 3.3.1

Voltage. Consider again the voltage X from Example 3.2.5. The distribution of X is characterized by the p.d.f. in Eq. (3.2.8). An alternative characterization that is more directly related to probabilities associated with X is obtained from the following function:

$$\begin{aligned}
F(x) = \Pr(X \leq x) &= \int_{-\infty}^x f(y)dy = \begin{cases} 0 & \text{for } x \leq 0, \\ \int_0^x \frac{dy}{(1+y)^2} & \text{for } x > 0, \end{cases} \\
&= \begin{cases} 0 & \text{for } x \leq 0, \\ 1 - \frac{1}{1+x} & \text{for } x > 0. \end{cases}
\end{aligned} \tag{3.3.1}$$

So, for example, $\Pr(X \leq 3) = F(3) = 3/4$. ◀

Definition and Basic Properties

Definition 3.3.1 (Cumulative) Distribution Function. The *distribution function* or *cumulative distribution function* (abbreviated *c.d.f.*) F of a random variable X is the function

$$F(x) = \Pr(X \leq x) \quad \text{for } -\infty < x < \infty. \tag{3.3.2}$$

It should be emphasized that the cumulative distribution function is defined as above for every random variable X , regardless of whether the distribution of X is discrete, continuous, or mixed. For the continuous random variable in Example 3.3.1, the c.d.f. was calculated in Eq. (3.3.1). Here is a discrete example:

Example 3.3.2

Bernoulli c.d.f. Let X have the Bernoulli distribution with parameter p defined in Definition 3.1.5. Then $\Pr(X = 0) = 1 - p$ and $\Pr(X = 1) = p$. Let F be the c.d.f. of X . It is easy to see that $F(x) = 0$ for $x < 0$ because $X \geq 0$ for sure. Similarly, $F(x) = 1$ for $x \geq 1$ because $X \leq 1$ for sure. For $0 \leq x < 1$, $\Pr(X \leq x) = \Pr(X = 0) = 1 - p$ because 0 is the only possible value of X that is in the interval $(-\infty, x]$. In summary,

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ 1 - p & \text{for } 0 \leq x < 1, \\ 1 & \text{for } x \geq 1. \end{cases} \quad \blacktriangleleft$$

We shall soon see (Theorem 3.3.2) that the c.d.f. allows calculation of all interval probabilities; hence, it characterizes the distribution of a random variable. It follows from Eq. (3.3.2) that the c.d.f. of each random variable X is a function F defined on the real line. The value of F at every point x must be a number $F(x)$ in the interval $[0, 1]$ because $F(x)$ is the probability of the event $\{X \leq x\}$. Furthermore, it follows from Eq. (3.3.2) that the c.d.f. of every random variable X must have the following three properties.

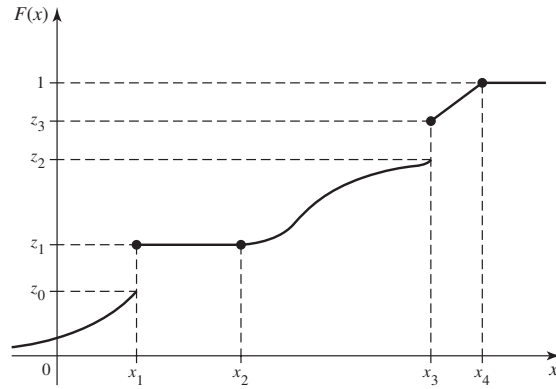
Property 3.3.1 **Nondecreasing.** The function $F(x)$ is nondecreasing as x increases; that is, if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$.

Proof If $x_1 < x_2$, then the event $\{X \leq x_1\}$ is a subset of the event $\{X \leq x_2\}$. Hence, $\Pr\{X \leq x_1\} \leq \Pr\{X \leq x_2\}$ according to Theorem 1.5.4. ■

An example of a c.d.f. is sketched in Fig. 3.6. It is shown in that figure that $0 \leq F(x) \leq 1$ over the entire real line. Also, $F(x)$ is always nondecreasing as x increases, although $F(x)$ is constant over the interval $x_1 \leq x \leq x_2$ and for $x \geq x_4$.

Property 3.3.2 **Limits at $\pm\infty$.** $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

Proof As in the proof of Property 3.3.1, note that $\{X \leq x_1\} \subset \{X \leq x_2\}$ whenever $x_1 < x_2$. The fact that $\Pr(X \leq x)$ approaches 0 as $x \rightarrow -\infty$ now follows from Exercise 13 in

Figure 3.6 An example of a c.d.f.

Section 1.10. Similarly, the fact that $\Pr(X \leq x)$ approaches 1 as $x \rightarrow \infty$ follows from Exercise 12 in Sec. 1.10. ■

The limiting values specified in Property 3.3.2 are indicated in Fig. 3.6. In this figure, the value of $F(x)$ actually becomes 1 at $x = x_4$ and then remains 1 for $x > x_4$. Hence, it may be concluded that $\Pr(X \leq x_4) = 1$ and $\Pr(X > x_4) = 0$. On the other hand, according to the sketch in Fig. 3.6, the value of $F(x)$ approaches 0 as $x \rightarrow -\infty$, but does not actually become 0 at any finite point x . Therefore, for every finite value of x , no matter how small, $\Pr(X \leq x) > 0$.

A c.d.f. need not be continuous. In fact, the value of $F(x)$ may jump at any finite or countable number of points. In Fig. 3.6, for instance, such jumps or points of discontinuity occur where $x = x_1$ and $x = x_3$. For each fixed value x , we shall let $F(x^-)$ denote the limit of the values of $F(y)$ as y approaches x from the left, that is, as y approaches x through values smaller than x . In symbols,

$$F(x^-) = \lim_{\substack{y \rightarrow x \\ y < x}} F(y).$$

Similarly, we shall define $F(x^+)$ as the limit of the values of $F(y)$ as y approaches x from the right. Thus,

$$F(x^+) = \lim_{\substack{y \rightarrow x \\ y > x}} F(y).$$

If the c.d.f. is continuous at a given point x , then $F(x^-) = F(x^+) = F(x)$ at that point.

Property 3.3.3

Continuity from the Right. A c.d.f. is always continuous from the right; that is, $F(x) = F(x^+)$ at every point x .

Proof Let $y_1 > y_2 > \dots$ be a sequence of numbers that are decreasing such that $\lim_{n \rightarrow \infty} y_n = x$. Then the event $\{X \leq x\}$ is the intersection of all the events $\{X \leq y_n\}$ for $n = 1, 2, \dots$. Hence, by Exercise 13 of Sec. 1.10,

$$F(x) = \Pr(X \leq x) = \lim_{n \rightarrow \infty} \Pr(X \leq y_n) = F(x^+). \quad \blacksquare$$

It follows from Property 3.3.3 that at every point x at which a jump occurs,

$$F(x^+) = F(x) \text{ and } F(x^-) < F(x).$$

In Fig. 3.6 this property is illustrated by the fact that, at the points of discontinuity $x = x_1$ and $x = x_3$, the value of $F(x_1)$ is taken as z_1 and the value of $F(x_3)$ is taken as z_3 .

Determining Probabilities from the Distribution Function

Example 3.3.3

Voltage. In Example 3.3.1, suppose that we want to know the probability that X lies in the interval $[2, 4]$. That is, we want $\Pr(2 \leq X \leq 4)$. The c.d.f. allows us to compute $\Pr(X \leq 4)$ and $\Pr(X \leq 2)$. These are related to the probability that we want as follows: Let $A = \{2 < X \leq 4\}$, $B = \{X \leq 2\}$, and $C = \{X \leq 4\}$. Because X has a continuous distribution, $\Pr(A)$ is the same as the probability that we desire. We see that $A \cup B = C$, and it is clear that A and B are disjoint. Hence, $\Pr(A) + \Pr(B) = \Pr(C)$. It follows that

$$\Pr(A) = \Pr(C) - \Pr(B) = F(4) - F(2) = \frac{4}{5} - \frac{3}{4} = \frac{1}{20}. \quad \blacktriangleleft$$

The type of reasoning used in Example 3.3.3 can be extended to find the probability that an arbitrary random variable X will lie in any specified interval of the real line from the c.d.f. We shall derive this probability for four different types of intervals.

Theorem 3.3.1

For every value x ,

$$\Pr(X > x) = 1 - F(x). \quad (3.3.3)$$

Proof The events $\{X > x\}$ and $\{X \leq x\}$ are disjoint, and their union is the whole sample space S whose probability is 1. Hence, $\Pr(X > x) + \Pr(X \leq x) = 1$. Now, Eq. (3.3.3) follows from Eq. (3.3.2). ■

Theorem 3.3.2

For all values x_1 and x_2 such that $x_1 < x_2$,

$$\Pr(x_1 < X \leq x_2) = F(x_2) - F(x_1). \quad (3.3.4)$$

Proof Let $A = \{x_1 < X \leq x_2\}$, $B = \{X \leq x_1\}$, and $C = \{X \leq x_2\}$. As in Example 3.3.3, A and B are disjoint, and their union is C , so

$$\Pr(x_1 < X \leq x_2) + \Pr(X \leq x_1) = \Pr(X \leq x_2).$$

Subtracting $\Pr(X \leq x_1)$ from both sides of this equation and applying Eq. (3.3.2) yields Eq. (3.3.4). ■

For example, if the c.d.f. of X is as sketched in Fig. 3.6, then it follows from Theorems 3.3.1 and 3.3.2 that $\Pr(X > x_2) = 1 - z_1$ and $\Pr(x_2 < X \leq x_3) = z_3 - z_1$. Also, since $F(x)$ is constant over the interval $x_1 \leq x \leq x_2$, then $\Pr(x_1 < X \leq x_2) = 0$.

It is important to distinguish carefully between the strict inequalities and the weak inequalities that appear in all of the preceding relations and also in the next theorem. If there is a jump in $F(x)$ at a given value x , then the values of $\Pr(X \leq x)$ and $\Pr(X < x)$ will be different.

Theorem 3.3.3

For each value x ,

$$\Pr(X < x) = F(x^-). \quad (3.3.5)$$

Proof Let $y_1 < y_2 < \cdots$ be an increasing sequence of numbers such that $\lim_{n \rightarrow \infty} y_n = x$. Then it can be shown that

$$\{X < x\} = \bigcup_{n=1}^{\infty} \{X \leq y_n\}.$$

Therefore, it follows from Exercise 12 of Sec. 1.10 that

$$\begin{aligned} \Pr(X < x) &= \lim_{n \rightarrow \infty} \Pr(X \leq y_n) \\ &= \lim_{n \rightarrow \infty} F(y_n) = F(x^-). \end{aligned} \quad \blacksquare$$

For example, for the c.d.f. sketched in Fig. 3.6, $\Pr(X < x_3) = z_2$ and $\Pr(X < x_4) = 1$.

Finally, we shall show that for every value x , $\Pr(X = x)$ is equal to the amount of the jump that occurs in F at the point x . If F is continuous at the point x , that is, if there is no jump in F at x , then $\Pr(X = x) = 0$.

Theorem 3.3.4 For every value x ,

$$\Pr(X = x) = F(x) - F(x^-). \quad (3.3.6)$$

Proof It is always true that $\Pr(X = x) = \Pr(X \leq x) - \Pr(X < x)$. The relation (3.3.6) follows from the fact that $\Pr(X \leq x) = F(x)$ at every point and from Theorem 3.3.3. $\blacksquare$

In Fig. 3.6, for example, $\Pr(X = x_1) = z_1 - z_0$, $\Pr(X = x_3) = z_3 - z_2$, and the probability of every other individual value of X is 0.

The c.d.f. of a Discrete Distribution

From the definition and properties of a c.d.f. $F(x)$, it follows that if $a < b$ and if $\Pr(a < X < b) = 0$, then $F(x)$ will be constant and horizontal over the interval $a < x < b$. Furthermore, as we have just seen, at every point x such that $\Pr(X = x) > 0$, the c.d.f. will jump by the amount $\Pr(X = x)$.

Suppose that X has a discrete distribution with the p.f. $f(x)$. Together, the properties of a c.d.f. imply that $F(x)$ must have the following form: $F(x)$ will have a jump of magnitude $f(x_i)$ at each possible value x_i of X , and $F(x)$ will be constant between every pair of successive jumps. The distribution of a discrete random variable X can be represented equally well by either the p.f. or the c.d.f. of X .

The c.d.f. of a Continuous Distribution

Theorem 3.3.5 Let X have a continuous distribution, and let $f(x)$ and $F(x)$ denote its p.d.f. and the c.d.f., respectively. Then F is continuous at every x ,

$$F(x) = \int_{-\infty}^x f(t) dt, \quad (3.3.7)$$

and

$$\frac{dF(x)}{dx} = f(x), \quad (3.3.8)$$

at all x such that f is continuous.

Proof Since the probability of each individual point x is 0, the c.d.f. $F(x)$ will have no jumps. Hence, $F(x)$ will be a continuous function over the entire real line.

By definition, $F(x) = \Pr(X \leq x)$. Since f is the p.d.f. of X , we have from the definition of p.d.f. that $\Pr(X \leq x)$ is the right-hand side of Eq. (3.3.7).

It follows from Eq. (3.3.7) and the relation between integrals and derivatives (the fundamental theorem of calculus) that, for every x at which f is continuous, Eq. (3.3.8) holds. ■

Thus, the c.d.f. of a continuous random variable X can be obtained from the p.d.f. and vice versa. Eq. (3.3.7) is how we found the c.d.f. in Example 3.3.1. Notice that the derivative of the F in Example 3.3.1 is

$$F'(x) = \begin{cases} 0 & \text{for } x < 0, \\ \frac{1}{(1+x)^2} & \text{for } x > 0, \end{cases}$$

and F' does not exist at $x = 0$. This verifies Eq. (3.3.8) for Example 3.3.1. Here, we have used the popular shorthand notation $F'(x)$ for the derivative of F at the point x .

Example 3.3.4

Calculating a p.d.f. from a c.d.f. Let the c.d.f. of a random variable be

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ x^{2/3} & \text{for } 0 \leq x \leq 1, \\ 1 & \text{for } x > 1. \end{cases}$$

This function clearly satisfies the three properties required of every c.d.f., as given earlier in this section. Furthermore, since this c.d.f. is continuous over the entire real line and is differentiable at every point except $x = 0$ and $x = 1$, the distribution of X is continuous. Therefore, the p.d.f. of X can be found at every point other than $x = 0$ and $x = 1$ by the relation (3.3.8). The value of $f(x)$ at the points $x = 0$ and $x = 1$ can be assigned arbitrarily. When the derivative $F'(x)$ is calculated, it is found that $f(x)$ is as given by Eq. (3.2.9) in Example 3.2.6. Conversely, if the p.d.f. of X is given by Eq. (3.2.9), then by using Eq. (3.3.7) it is found that $F(x)$ is as given in this example. ◀

The Quantile Function

Example 3.3.5

Fair Bets. Suppose that X is the amount of rain that will fall tomorrow, and X has c.d.f. F . Suppose that we want to place an even-money bet on X as follows: If $X \leq x_0$, we win one dollar and if $X > x_0$ we lose one dollar. In order to make this bet fair, we need $\Pr(X \leq x_0) = \Pr(X > x_0) = 1/2$. We could search through all of the real numbers x trying to find one such that $F(x) = 1/2$, and then we would let x_0 equal the value we found. If F is a one-to-one function, then F has an inverse F^{-1} and $x_0 = F^{-1}(1/2)$. ◀

The value x_0 that we seek in Example 3.3.5 is called the 0.5 *quantile* of X or the 50th *percentile* of X because 50% of the distribution of X is at or below x_0 .

Definition 3.3.2

Quantiles/Percentiles. Let X be a random variable with c.d.f. F . For each p strictly between 0 and 1, define $F^{-1}(p)$ to be the smallest value x such that $F(x) \geq p$. Then $F^{-1}(p)$ is called the p *quantile* of X or the 100

percentile of X . The function F^{-1} defined here on the open interval $(0, 1)$ is called the *quantile function* of X .

Example
3.3.6

Standardized Test Scores. Many universities in the United States rely on standardized test scores as part of their admissions process. Thousands of people take these tests each time that they are offered. Each examinee's score is compared to the collection of scores of all examinees to see where it fits in the overall ranking. For example, if 83% of all test scores are at or below your score, your test report will say that you scored at the 83rd percentile. ◀

The notation $F^{-1}(p)$ in Definition 3.3.2 deserves some justification. Suppose first that the c.d.f. F of X is continuous and one-to-one over the whole set of possible values of X . Then the inverse F^{-1} of F exists, and for each $0 < p < 1$, there is one and only one x such that $F(x) = p$. That x is $F^{-1}(p)$. Definition 3.3.2 extends the concept of inverse function to nondecreasing functions (such as c.d.f.'s) that may be neither one-to-one nor continuous.

Quantiles of Continuous Distributions When the c.d.f. of a random variable X is continuous and one-to-one over the whole set of possible values of X , the inverse F^{-1} of F exists and equals the quantile function of X .

Example
3.3.7

Value at Risk. The manager of an investment portfolio is interested in how much money the portfolio might lose over a fixed time horizon. Let X be the change in value of the given portfolio over a period of one month. Suppose that X has the p.d.f. in Fig. 3.7. The manager computes a quantity known in the world of risk management as *Value at Risk* (denoted by VaR). To be specific, let $Y = -X$ stand for the loss incurred by the portfolio over the one month. The manager wants to have a level of confidence about how large Y might be. In this example, the manager specifies a probability level, such as 0.99 and then finds y_0 , the 0.99 quantile of Y . The manager is now 99% sure that $Y \leq y_0$, and y_0 is called the VaR. If X has a continuous distribution, then it is easy to see that y_0 is closely related to the 0.01 quantile of the distribution of X . The 0.01 quantile x_0 has the property that $\Pr(X < x_0) = 0.01$. But $\Pr(X < x_0) = \Pr(Y > -x_0) = 1 - \Pr(Y \leq -x_0)$. Hence, $-x_0$ is a 0.99 quantile of Y . For the p.d.f. in Fig. 3.7, we see that $x_0 = -4.14$, as the shaded region indicates. Then $y_0 = 4.14$ is VaR for one month at probability level 0.99. ◀

Figure 3.7 The p.d.f. of the change in value of a portfolio with lower 1% indicated.

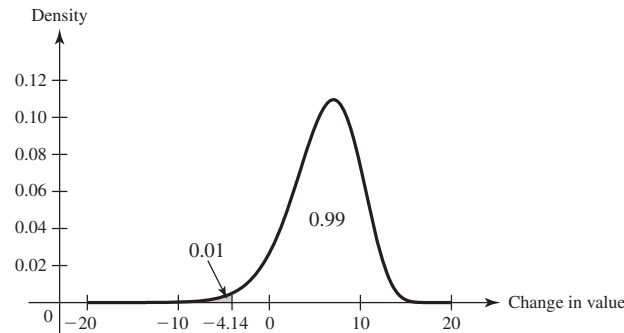
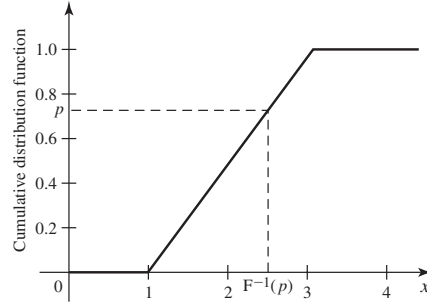


Figure 3.8 The c.d.f. of a uniform distribution indicating how to solve for a quantile.



Example 3.3.8

Uniform Distribution on an Interval. Let X have the uniform distribution on the interval $[a, b]$. The c.d.f. of X is

$$F(x) = \Pr(X \leq x) = \begin{cases} 0 & \text{if } x \leq a, \\ \int_a^x \frac{1}{b-a} du & \text{if } a < x \leq b, \\ 1 & \text{if } x > b. \end{cases}$$

The integral above equals $(x-a)/(b-a)$. So, $F(x) = (x-a)/(b-a)$ for all $a < x < b$, which is a strictly increasing function over the entire interval of possible values of X . The inverse of this function is the quantile function of X , which we obtain by setting $F(x)$ equal to p and solving for x :

$$\begin{aligned} \frac{x-a}{b-a} &= p, \\ x-a &= p(b-a), \\ x &= a + p(b-a) = pb + (1-p)a. \end{aligned}$$

Figure 3.8 illustrates how the calculation of a quantile relates to the c.d.f.

The quantile function of X is $F^{-1}(p) = pb + (1-p)a$ for $0 < p < 1$. In particular, $F^{-1}(1/2) = (b+a)/2$. ◀

Note: Quantiles, Like c.d.f.'s, Depend on the Distribution Only. Any two random variables with the same distribution have the same quantile function. When we refer to a quantile of X , we mean a quantile of the distribution of X .

Quantiles of Discrete Distributions It is convenient to be able to calculate quantiles for discrete distributions as well. The quantile function of Definition 3.3.2 exists for all distributions whether discrete, continuous, or otherwise. For example, in Fig. 3.6, let $z_0 \leq p \leq z_1$. Then the smallest x such that $F(x) \geq p$ is x_1 . For every value of $x < x_1$, we have $F(x) < z_0 \leq p$ and $F(x_1) = z_1$. Notice that $F(x) = z_1$ for all x between x_1 and x_2 , but since x_1 is the smallest of all those numbers, x_1 is the p quantile. Because distribution functions are continuous from the right, the smallest x such that $F(x) \geq p$ exists for all $0 < p < 1$. For $p = 1$, there is no guarantee that such an x will exist. For example, in Fig. 3.6, $F(x_4) = 1$, but in Example 3.3.1, $F(x) < 1$ for all x . For $p = 0$, there is never a smallest x such that $F(x) = 0$ because $\lim_{x \rightarrow -\infty} F(x) = 0$. That is, if $F(x_0) = 0$, then $F(x) = 0$ for all $x < x_0$. For these reasons, we never talk about the 0 or 1 quantiles.

Table 3.1 Quantile function for Example 3.3.9

p	$F^{-1}(p)$
$(0, 0.1681]$	0
$(0.1681, 0.5283]$	1
$(0.5283, 0.8370]$	2
$(0.8370, 0.9693]$	3
$(0.9693, 0.9977]$	4
$(0.9977, 1)$	5

Example 3.3.9

Quantiles of a Binomial Distribution. Let X have the binomial distribution with parameters 5 and 0.3. The binomial table in the back of the book has the p.f. f of X , which we reproduce here together with the c.d.f. F :

x	0	1	2	3	4	5
$f(x)$	0.1681	0.3602	0.3087	0.1323	0.0284	0.0024
$F(x)$	0.1681	0.5283	0.8370	0.9693	0.9977	1

(A little rounding error occurred in the p.f.) So, for example, the 0.5 quantile of this distribution is 1, which is also the 0.25 quantile and the 0.20 quantile. The entire quantile function is in Table 3.1. So, the 90th percentile is 3, which is also the 95th percentile, etc. ◀

Certain quantiles have special names.

Definition 3.3.3

Median/Quartiles. The $1/2$ quantile or the 50th percentile of a distribution is called its *median*. The $1/4$ quantile or 25th percentile is the *lower quartile*. The $3/4$ quantile or 75th percentile is called the *upper quartile*.

Note: The Median Is Special. The median of a distribution is one of several special features that people like to use when summarizing the distribution of a random variable. We shall discuss summaries of distributions in more detail in Chapter 4. Because the median is such a popular summary, we need to note that there are several different but similar “definitions” of median. Recall that the $1/2$ quantile is the *smallest* number x such that $F(x) \geq 1/2$. For some distributions, usually discrete distributions, there will be an interval of numbers $[x_1, x_2]$ such that for all $x \in [x_1, x_2]$, $F(x) = 1/2$. In such cases, it is common to refer to all such x (including x_2) as medians of the distribution. (See Definition 4.5.1.) Another popular convention is to call $(x_1 + x_2)/2$ the median. This last is probably the most common convention. The readers should be aware that, whenever they encounter a median, it might be any one of the things that we just discussed. Fortunately, they all mean nearly the same thing, namely that the number divides the distribution in half as closely as is possible.

Example
3.3.10

Uniform Distribution on Integers. Let X have the uniform distribution on the integers 1, 2, 3, 4. (See Definition 3.1.6.) The c.d.f. of X is

$$F(x) = \begin{cases} 0 & \text{if } x < 1, \\ 1/4 & \text{if } 1 \leq x < 2, \\ 1/2 & \text{if } 2 \leq x < 3, \\ 3/4 & \text{if } 3 \leq x < 4, \\ 1 & \text{if } x \geq 4. \end{cases}$$

The 1/2 quantile is 2, but every number in the interval $[2, 3]$ might be called a median. The most popular choice would be 2.5. ◀

One advantage to describing a distribution by the quantile function rather than by the c.d.f. is that quantile functions are easier to display in tabular form for multiple distributions. The reason is that the domain of the quantile function is always the interval $(0, 1)$ no matter what the possible values of X are. Quantiles are also useful for summarizing distributions in terms of where the probability is. For example, if one wishes to say where the middle half of a distribution is, one can say that it lies between the 0.25 quantile and the 0.75 quantile. In Sec. 8.5, we shall see how to use quantiles to help provide estimates of unknown quantities after observing data.

In Exercise 19, you can show how to recover the c.d.f. from the quantile function. Hence, the quantile function is an alternative way to characterize a distribution.

Summary

The c.d.f. F of a random variable X is $F(x) = \Pr(X \leq x)$ for all real x . This function is continuous from the right. If we let $F(x^-)$ equal the limit of $F(y)$ as y approaches x from below, then $F(x) - F(x^-) = \Pr(X = x)$. A continuous distribution has a continuous c.d.f. and $F'(x) = f(x)$, the p.d.f. of the distribution, for all x at which F is differentiable. A discrete distribution has a c.d.f. that is constant between the possible values and jumps by $f(x)$ at each possible value x . The quantile function $F^{-1}(p)$ is equal to the smallest x such that $F(x) \geq p$ for $0 < p < 1$.

Exercises

1. Suppose that a random variable X has the Bernoulli distribution with parameter $p = 0.7$. (See Definition 3.1.5.) Sketch the c.d.f. of X .

2. Suppose that a random variable X can take only the values $-2, 0, 1$, and 4 , and that the probabilities of these values are as follows: $\Pr(X = -2) = 0.4$, $\Pr(X = 0) = 0.1$, $\Pr(X = 1) = 0.3$, and $\Pr(X = 4) = 0.2$. Sketch the c.d.f. of X .

3. Suppose that a coin is tossed repeatedly until a head is obtained for the first time, and let X denote the number of tosses that are required. Sketch the c.d.f. of X .

4. Suppose that the c.d.f. F of a random variable X is as sketched in Fig. 3.9. Find each of the following probabilities:

- | | |
|---------------------------|---------------------------|
| a. $\Pr(X = -1)$ | b. $\Pr(X < 0)$ |
| c. $\Pr(X \leq 0)$ | d. $\Pr(X = 1)$ |
| e. $\Pr(0 < X \leq 3)$ | f. $\Pr(0 < X < 3)$ |
| g. $\Pr(0 \leq X \leq 3)$ | h. $\Pr(1 < X \leq 2)$ |
| i. $\Pr(1 \leq X \leq 2)$ | j. $\Pr(X > 5)$ |
| k. $\Pr(X \geq 5)$ | l. $\Pr(3 \leq X \leq 4)$ |

5. Suppose that the c.d.f. of a random variable X is as follows:

$$F(x) = \begin{cases} 0 & \text{for } x \leq 0, \\ \frac{1}{9}x^2 & \text{for } 0 < x \leq 3, \\ 1 & \text{for } x > 3. \end{cases}$$

Find and sketch the p.d.f. of X .

6. Suppose that the c.d.f. of a random variable X is as follows:

$$F(x) = \begin{cases} e^{x-3} & \text{for } x \leq 3, \\ 1 & \text{for } x > 3. \end{cases}$$

Find and sketch the p.d.f. of X .

7. Suppose, as in Exercise 7 of Sec. 3.2, that a random variable X has the uniform distribution on the interval $[-2, 8]$. Find and sketch the c.d.f. of X .

8. Suppose that a point in the xy -plane is chosen at random from the interior of a circle for which the equation is $x^2 + y^2 = 1$; and suppose that the probability that the point will belong to each region inside the circle is proportional to the area of that region. Let Z denote a random variable representing the distance from the center of the circle to the point. Find and sketch the c.d.f. of Z .

9. Suppose that X has the uniform distribution on the interval $[0, 5]$ and that the random variable Y is defined by $Y = 0$ if $X \leq 1$, $Y = 5$ if $X \geq 3$, and $Y = X$ otherwise. Sketch the c.d.f. of Y .

10. For the c.d.f. in Example 3.3.4, find the quantile function.

11. For the c.d.f. in Exercise 5, find the quantile function.

12. For the c.d.f. in Exercise 6, find the quantile function.

13. Suppose that a broker believes that the change in value X of a particular investment over the next two months has the uniform distribution on the interval $[-12, 24]$. Find the value at risk VaR for two months at probability level 0.95.

14. Find the quartiles and the median of the binomial distribution with parameters $n = 10$ and $p = 0.2$.

15. Suppose that X has the p.d.f.

$$f(x) = \begin{cases} 2x & \text{if } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find and sketch the c.d.f. or X .

16. Find the quantile function for the distribution in Example 3.3.1.

17. Prove that the quantile function F^{-1} of a general random variable X has the following three properties that are analogous to properties of the c.d.f.:

- F^{-1} is a nondecreasing function of p for $0 < p < 1$.
- Let $x_0 = \lim_{p \rightarrow 0} F^{-1}(p)$ and $x_1 = \lim_{p \rightarrow 1} F^{-1}(p)$. Then x_0 equals the greatest lower bound on the set of numbers c such that $\Pr(X \leq c) > 0$, and x_1 equals the least upper bound on the set of numbers d such that $\Pr(X \geq d) > 0$.
- F^{-1} is continuous from the left; that is $F^{-1}(p) = F^{-1}(p^-)$ for all $0 < p < 1$.

18. Let X be a random variable with quantile function F^{-1} . Assume the following three conditions: (i) $F^{-1}(p) = c$ for all p in the interval (p_0, p_1) , (ii) either $p_0 = 0$ or $F^{-1}(p_0) < c$, and (iii) either $p_1 = 1$ or $F^{-1}(p) > c$ for $p > p_1$. Prove that $\Pr(X = c) = p_1 - p_0$.

19. Let X be a random variable with c.d.f. F and quantile function F^{-1} . Let x_0 and x_1 be as defined in Exercise 17. (Note that $x_0 = -\infty$ and/or $x_1 = \infty$ are possible.) Prove that for all x in the open interval (x_0, x_1) , $F(x)$ is the largest p such that $F^{-1}(p) \leq x$.

20. In Exercise 13 of Sec. 3.2, draw a sketch of the c.d.f. F of X and find $F(10)$.

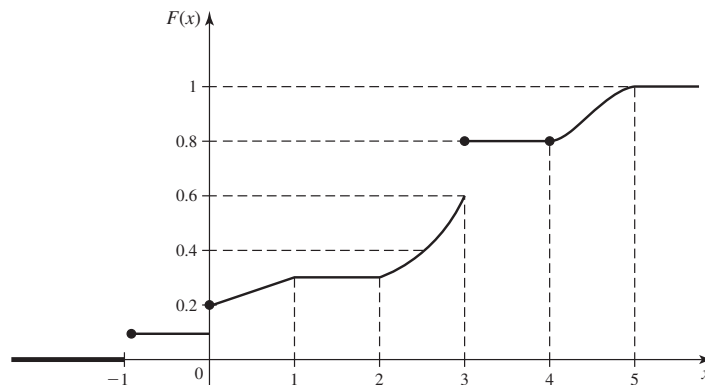


Figure 3.9 The c.d.f. for Exercise 4.